CS 486/686
Reinforcement Learning

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Lecture 15

RN 22.1–22.3 · PM 12.1, 12.5, 12.8

Heads-up: next week is asynchronous

🎥 I'll be away at **ICML** the week of July 7. **L17 (Tue Jul 7)** and **L18 (Thu Jul 9)** will be **pre-recorded videos** — no in-person class those days. Watch on your own time (links posted before class).

📅 **Deadlines are unchanged:** Chat 8 + the CS686 project proposal are due **Tue Jul 7**; Chat 9 is out and **Assignment 2 is due Thu Jul 9**.

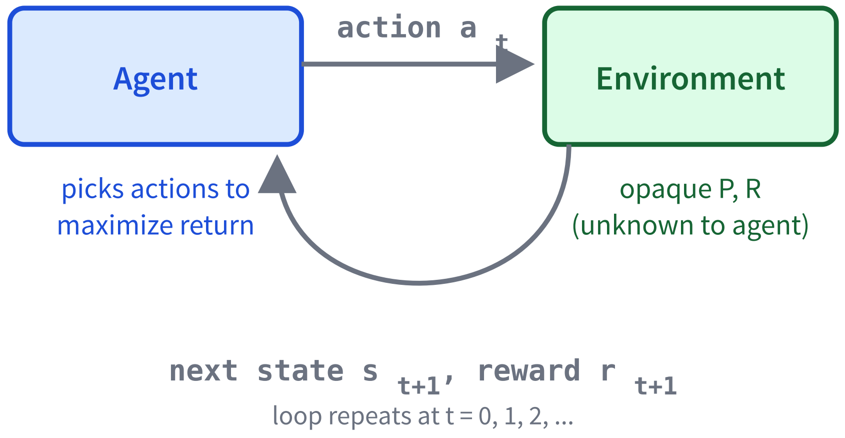
💬 Questions? Post on **Piazza** — the TAs are available all week.

Search › Uncertainty › **Decisions** › Learning

Learning goals

- Trace + implement **passive adaptive dynamic programming** (passive ADP).
- Explain the **exploration vs exploitation** trade-off.
- Trace + implement **active ADP**.
- Trace + implement **Q-learning**.
- Trace + implement **SARSA**.
- Tell **on-policy** from **off-policy** learning.

The RL setting: no P, no R up front



An MDP whose **transition** P and **reward** R are unknown. The agent sees state s and reward r, picks an action a, and observes the next state s'.

Goal: maximize discounted return — while learning the world as it goes.

Passive learning: just evaluate a fixed policy

Fix a policy π; goal: learn $V^{\pi}(s)$ for every state s. **The catch:** we no longer know the reward $R(s)$ or the transitions $P(s' \mid s, a)$ — we only get to *observe* experiences.

Adaptive Dynamic Programming (ADP)

1. **Learn** the transition $P(s' \mid s, a)$ and reward $R(s)$ from observed experience.
2. **Solve** the Bellman policy-evaluation equations:

$$V^{\pi}(s) = R(s) + \gamma \textstyle\sum_{s'} P(s' \mid s, \pi(s))\, V^{\pi}(s').$$

A **model-based** approach — it builds an explicit model of the world.

The passive ADP algorithm

`passive-ADP(`π`)`

1. Initialize $R(s),\ N(s,a),\ N(s,a,s'),\ V^\pi(s)$.
2. Follow π and observe an experience $\langle s, a, s', r \rangle$.
3. Update reward: $R(s) \leftarrow r$.
4. Update counts and transition probability:

$$N(s,a) \mathrel{+}= 1, \quad N(s,a,s') \mathrel{+}= 1, \quad P(s' \mid s,a) = \tfrac{N(s,a,s')}{N(s,a)}.$$

5. Re-solve Bellman to get $V^\pi(s)$ (exactly or iteratively).

Passive ADP: update the model

Same grid world as before (the agent moves in the intended direction, or slips to either side) — but now **we don't know** those slip probabilities, so we **count outcomes** to estimate P.

s_{11}	+1
s_{21}	−1

$\pi(s_{11}) = \mathrm{down}, \pi(s_{21}) = \mathrm{right}, \gamma = 0.9.$

Counts so far: $N(s,a) = 5$, with $N(s,a,s') = 3$ for the intended direction and 1 for each side.

New experience: $\langle s_{11}, \mathrm{down}, s_{21}, -0.04 \rangle$.

Increment counts: $N(s_{11}, \mathrm{down})$: $5 \to 6$, $N(s_{11}, \mathrm{down}, s_{21})$: $3 \to 4$.

Re-estimate $P(\cdot \mid s_{11}, \mathrm{down})$ as count fractions:

$$s_{21}: \tfrac{4}{6}, \quad \text{stay } s_{11}: \tfrac{1}{6}, \quad +1: \tfrac{1}{6}$$

(Going down can slip sideways: left bumps the wall → stay; right → the $+1$ cell.)

Passive ADP: re-solve with the new model

Reminder — policy evaluation: $V^\pi(s) = R(s) + \gamma \sum_{s'} P(s' \mid s, \pi(s))\, V^\pi(s')$ ($R = -0.04, \gamma = 0.9$).

Estimate $P(\cdot \mid s_{21}, \mathrm{right})$ the same way (counts 3 intended, 1 each side, out of 5): right $\to -1$ is $\frac{3}{5}$; up $\to s_{11}$ is $\frac{1}{5}$; down bumps the wall $\to$ stay s_{21} is $\frac{1}{5}$.

$$V(s_{11}) = -0.04 + 0.9\left[\tfrac{4}{6}V(s_{21}) + \tfrac{1}{6}V(s_{11}) + \tfrac{1}{6}(+1)\right]$$
$$V(s_{21}) = -0.04 + 0.9\left[\tfrac{3}{5}(-1) + \tfrac{1}{5}V(s_{11}) + \tfrac{1}{5}V(s_{21})\right]$$

Two linear equations, two unknowns — solve:

$$V(s_{11}) = -0.438, \quad V(s_{21}) = -0.803.$$

Quick check: probability update

Q1. In a 3x4 grid, suppose $N(s_{23}, \text{down}) = 23$ and $N(s_{23}, \text{down}, s_{24}) = 2$. The agent tries to move *down* but slips and lands in s_{24}. What is the updated $P(s_{24} \mid s_{23}, \text{down})$?

A. $2/23 \approx 0.087$

B. $3/23 \approx 0.130$

C. $3/24 = 0.125$

D. $2/24 \approx 0.083$

C — 0.125. First increment the counts: $N(s_{23}, \text{down}) \to 24$ and $N(s_{23}, \text{down}, s_{24}) \to 3$. Then $P = 3/24 = 0.125$.

Active learning: explore vs exploit

Now the agent *chooses* its actions. Two competing pressures:

Exploit

Take the action that maximizes the current $V(s)$ (or $Q(s, a)$). Reap known rewards.

Explore

Try a different action. Gather data to learn a better model / better Q-values.

> A purely greedy agent *seldom* converges to the optimal policy — the learned model is biased toward states the agent has already preferred.

Exploration 1: ε-greedy

action probability

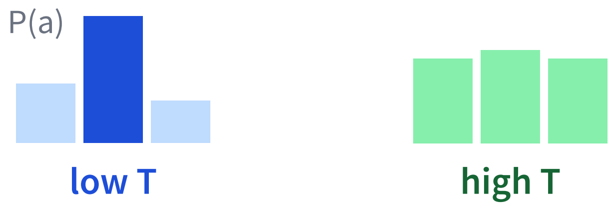
With probability ε: take a **random** action (explore).

With probability $1 - \varepsilon$: take the **greedy** action (exploit).

Start with a large ε and **decay** it over time — explore early, exploit once the values settle.

Simple and popular, but it treats every sub-optimal action as equally worth trying.

Exploration 2: softmax (Boltzmann)



Pick actions in proportion to their value, tuned by a **temperature** T:

$$P(a) = \frac{e^{Q(s,a)/T}}{\sum_{a'} e^{Q(s,a')/T}}$$

- Low T: nearly **greedy** (peaked on the best action).
- High T: nearly **uniform** (lots of exploration).

Unlike ε-greedy, better actions are explored more than clearly-bad ones. Cool T over time (like simulated annealing).

Exploration 3: optimistic initialization

Assume the best about actions you haven't tried enough. Replace a value u with an **exploration function** f:

$$f(u,n) = \begin{cases} R^+ & n < N_e \\ u & \text{otherwise} \end{cases}$$

R^+ is an optimistic reward bound; N_e is how many tries before we trust the real estimate u.

Every action looks great until tried N_e times, pulling the agent toward the unknown.

Active ADP: bake exploration into the value

Active ADP = passive ADP, but now the agent *chooses* actions. The trick: act greedily on an **optimistic** value V^+, seen through the exploration function f from before.

Pick the action whose successor value — *viewed through* f — looks best:

$$a = \arg\max_a \ f\left(\underbrace{\textstyle\sum_{s'} P(s' \mid s, a)\, V^+(s')}_{\text{estimated value of } a},\ \ \underbrace{N(s,a)}_{\text{times tried}} \right)$$

f inflates the value of **under-tried** actions ($N(s,a) < N_e$), so the agent is pulled toward exploring them.

The active ADP algorithm

`active-ADP(`$N_e,\ R^+$`)`

1. Initialize $R(s),\ V^+(s),\ N(s,a),\ N(s,a,s')$.
2. Repeat until every (s,a) is tried $\geq N_e$ times and V^+ converges:
   - Pick $a = \arg\max_a f(\sum_{s'} P(s' \mid s,a)\, V^+(s'),\ N(s,a))$ (the rule above).
   - Execute a; observe $\langle s, a, s', r\rangle$; update $R(s) \leftarrow r$, counts, and $P(s' \mid s,a)$.
   - $V^+(s) \leftarrow R(s) + \gamma\ \max_a f(\sum_{s'} P(s' \mid s,a)\, V^+(s'),\ N(s,a))$.
   - $s \leftarrow s'$.

Same model-based loop as passive ADP — only the action choice and the $\max$ now pass through f.

Recall: the Bellman equation for $V^\star$

From L14, two connections between $V^\star$ and $Q^\star$:

$$① \; V^\star(s) = \max_a Q^\star(s,a)$$

$$② \; Q^\star(s,a) = R(s) + \gamma \sum_{s'} P(s' \mid s,a)\, V^\star(s')$$

Substitute ② into ① (replace each $Q^\star$ with reward + future value):

$$V^\star(s) = R(s) + \gamma \max_a \sum_{s'} P(s' \mid s,a)\, V^\star(s')$$

The Bellman equation for $Q^\star$

Now do the reverse — substitute ① $\big(V^\star(s') = \max_{a'} Q^\star(s', a')\big)$ into ②:

$$\begin{aligned} Q^\star(s,a) &= R(s) + \gamma \sum_{s'} P(s' \mid s,a)\, \underline{V^\star(s')} \\ &= R(s) + \gamma \sum_{s'} P(s' \mid s,a)\, \underline{\max_{a'} Q^\star(s',a')} \end{aligned}$$

Given $Q^\star$, the best action is $\pi^\star(s) = \arg\max_a Q^\star(s,a)$ — **no P needed.** That is what lets Q-learning be **model-free**.

Temporal difference (TD) error

After one transition $\langle s, a, r, s' \rangle$, pretend that transition was certain ($P(s' \mid s, a) = 1$). Then by Bellman, $Q(s, a)$ should equal the **target**:

$$Q(s,a) \overset{?}{=} \underbrace{R(s) + \gamma \max_{a'} Q(s', a')}_{\text{target}}$$

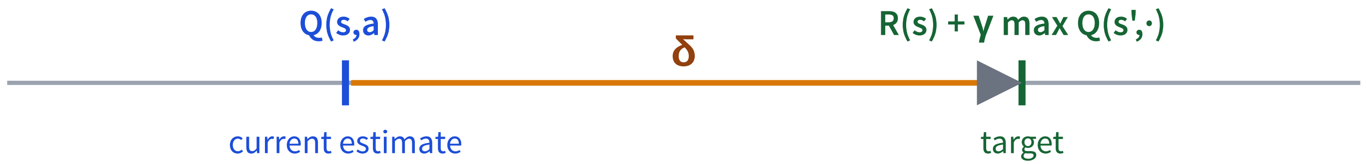
TD error

$$\delta = \left(R(s) + \gamma \max_{a'} Q(s', a')\right) - Q(s,a)$$

How far the current estimate $Q(s, a)$ is from the one-step target.

From the TD error to the update

One sample is noisy, so don't jump all the way — **nudge** $Q(s, a)$ toward the target by a fraction α of the gap δ:



$$Q(s,a) \;\leftarrow\; Q(s,a) + \alpha\,\delta$$
$$= \; Q(s,a) + \alpha\Big(\underbrace{R(s) + \gamma \max_{a'} Q(s',a')}_{\text{target}} - Q(s,a)\Big)$$

$\alpha \to 0$: barely move; $\alpha = 1$: jump straight to the target. This update rule *is* Q-learning.

The Q-learning algorithm

`Q-learning(`α, γ`)`

1. Initialize $Q(s, a)$ arbitrarily; observe the starting state s.
2. Repeat until Q converges:
   - Pick a from s using *any* exploration strategy (ε-greedy, softmax, optimistic).
   - Execute a; observe reward r and next state s'.
   - $Q(s,a) \leftarrow Q(s,a) + \alpha\big(r + \gamma\ \max_{a'} Q(s',a') - Q(s,a)\big)$.
   - $s \leftarrow s'$.

Model-free — it never estimates P; converges to $Q^\star$ if every (s, a) is visited infinitely often.

Choosing the learning rate α

The algorithm works for any fixed $\alpha \in (0, 1]$, but a good schedule lets α **shrink** as a state–action pair is visited more often:

$$\alpha(N) = \frac{10}{9 + N(s, a)}$$

- Early visits (small N): α is large — move Q a lot on fresh information.
- Later visits (large N): α is small — fine-tune and average out noise.
- A decaying α is exactly what guarantees convergence to $Q^\star$.

Q-learning: a single update

3x4 grid. Experience: $\langle s_{32}, \text{right}, s_{33}, -0.04\rangle$. Take $\gamma = 1, \alpha = 0.1$.

	s_{32}	s_{33}
up	0.55	0.4
right	0.7	0.9
down	0.5	0.8
left	0.4	0.5

Yellow cell = the one we update; green = the max-value action in s_{33}.

$$\textbf{target} = r + \gamma \cdot \max_{a'} Q(s_{33}, a') = -0.04 + 1.0 \cdot 0.9 = 0.86$$

$$\delta = \text{target} - Q(s_{32}, \text{right}) = 0.86 - 0.7 = 0.16$$

$$Q(s_{32}, \text{right}) \leftarrow 0.7 + 0.1 \cdot 0.16 = 0.716$$

SARSA: on-policy temporal difference

Observe a 5-tuple $\langle s, a, r, s', a' \rangle$ where a' is the action **actually taken** in s' (per the current policy):

Q-learning (off-policy)

$$Q(s,a) \leftarrow Q(s,a) + \alpha\big(r + \gamma \max_{a'} Q(s',a') - Q(s,a)\big)$$

Target uses the *greedy* next action. Learns about the optimal policy regardless of what's actually played.

SARSA (on-policy)

$$Q(s,a) \leftarrow Q(s,a) + \alpha\big(r + \gamma\, Q(s',a') - Q(s,a)\big)$$

Target uses the *actually-played* next action a'. Learns about the policy the agent is following (exploration and all).

SARSA = State, Action, Reward, State, Action.

Why is Q-learning *off*-policy?

Every step, the agent runs **two** policies at once:

- **Behavior policy** — how it *actually* picks actions (ε-greedy: usually greedy, sometimes explores).
- **Target policy** — the policy whose value the update is learning about.

Q-learning

The update uses $\max_{a'} Q(s', a')$ — the value of the **greedy** next action — even when the agent is about to *explore* a different one.

Behavior $\neq$ target $\Rightarrow$ **off-policy**. Converges to $Q^\star$ no matter how it explores.

SARSA

The update uses $Q(s', a')$ for the action **actually chosen** by ε-greedy — the very move it will play next.

Behavior $=$ target $\Rightarrow$ **on-policy**. Learns the value of the exploring policy itself.

Off-policy: learn about one policy (greedy) while *behaving* by another (exploratory). **On-policy:** learn about the policy you actually run.

SARSA vs Q-learning: the scenario

Update $Q(s_{13}, \text{down}) = 0.4$ after $\langle s_{13}, \text{down}, s_{23} \rangle$, with $r = -0.04,\ \gamma = 0.9,\ \alpha = 0.4.$

The ε-greedy step **explores** $a' = \text{right}$ in s_{23} (into the bad cell).

$Q(s_{23}, \cdot)$	up	right	down	left
	0.3	-0.4	0.7	0.3

Yellow = the explored next action $a' = \text{right}$ (SARSA's target); green = $\max_{a'} Q$ (Q-learning's target).

SARSA vs Q-learning: the two updates

SARSA: uses $a' = \text{right}$

```
target = −0.04 + 0.9·(−0.4) = −0.4
δ = −0.4 − 0.4 = −0.8
Q(s13, down) ← 0.4 + 0.4 · (−0.8)
```

= 0.08 (punished for the bad step)

Q-learning: uses $\max_{a'} = 0.7$

```
target = −0.04 + 0.9 · 0.7 = 0.59
δ = 0.59 − 0.4 = 0.19
Q(s13, down) ← 0.4 + 0.4 · 0.19
```

= 0.476 (rewarded for the best)

The agent is about to play $a' = \text{right}$ (-0.4), but Q-learning updates toward $\max = 0.7$ (down) — not the move it will make. **That mismatch is "off-policy."**

SARSA is **conservative** (learns what really happens); Q-learning is **aggressive** (learns the optimal policy).

When to use what (Q2)

Q2. For a high-stakes *online* learning problem (for example, a self-driving car still in training), which algorithm is safer?

A. Q-learning

B. **SARSA**

C. I don't know

B — SARSA. SARSA's target uses the action that will *actually* be taken while exploring, so it learns to avoid catastrophic outcomes along the exploration path. Q-learning's target uses the greedy action, ignoring the cost of exploration mistakes.

Putting the labels together (Q3)

Q3. For each of **Q-learning** and **SARSA**: is it *model-based* or *model-free*? Is it *on-policy* or *off-policy*?

> **Both are model-free** (neither estimates P). **Q-learning is off-policy** — its target uses $\max_{a'} Q(s', a')$, the greedy action. **SARSA is on-policy** — its target uses the action actually played.

ADP vs Q-learning vs SARSA

	ADP	Q-learning	SARSA
Learns transition P?	yes (model-based)	no (model-free)	no (model-free)
On / off policy	*n/a* (eval-only)	off-policy	on-policy
Computation per experience	heavy (solve Bellman)	cheap (one TD update)	cheap (one TD update)
Convergence speed	fast (few experiences)	slower, higher variance	slower, higher variance

ADP is sample-efficient but expensive per step; TD methods (Q, SARSA) flip that trade-off.

Learning goals (recap)

- ✓ Trace + implement **passive ADP**.
- ✓ Explain the **exploration vs exploitation** trade-off.
- ✓ Trace + implement **active ADP**.
- ✓ Trace + implement **Q-learning**.
- ✓ Trace + implement **SARSA.**
- ✓ Tell **on-policy** from **off-policy**.

Search › Uncertainty › Decisions › **Learning**

Next module: Machine Learning and Deep Learning

RL was learning from rewards. The last module turns to *learning from data*.

- **L16** — supervised learning, bias / variance, cross-validation.
- **L17** — unsupervised: k-means, PCA, autoencoders, GANs.
- **L18** — decision trees: entropy, information gain, ID3.
- **L19–L21** — neural networks: forward / backward / optimizers.